# CS 170 Assignment 6, TTT question

**5. (5 pts)** Write a class named `TTT` that generates a Tic-Tac-Toe board of size N (provided by the user via `Scanner`) filled randomly with 'X' and 'O'. The board is a two-dimensional array. Your program must satisfy the following requirements:

1. The 'X' and 'O' marks are assigned to board positions randomly.
2. The number of 'X' marks and 'O' marks must be as balanced as possible: the absolute difference between them must be at most 1. Specifically, when N is even, exactly $N \times N/2$ positions are filled with 'X' and the rest with 'O' (a difference of 0); when N is odd, the difference between the count of 'X' and the count of 'O' is exactly 1.

**6. (5 pts)** Write a method `boolean checkTTT(char[][] TTTboard)` that determines whether the board has exactly one winner. The method returns `true` if exactly one mark wins (it does not matter whether 'X' or 'O' wins), and returns `false` if there is no winner or if both 'X' and 'O' win. On a board of size N, 'X' (or 'O') wins if N consecutive identical marks appear in any full row, any full column, or either main diagonal. You may define additional helper methods and call them from within `checkTTT`.